Veganism and its relationship with insulin resistance and intramyocellular lipid.

OBJECTIVE: To test the hypothesis that dietary factors in the vegan diet lead to improved insulin sensitivity and lower intramyocellular lipid (IMCL) storage. DESIGN: Case-control study. SETTING: Imperial College School of Medicine, Hammersmith Hospital Campus, London, UK. SUBJECTS: A total of 24 vegans and 25 omnivores participated in this study; three vegan subjects could not be matched therefore the matched results are shown for 21 vegans and 25 omnivores. The subjects were matched for gender, age and body mass index (BMI). INTERVENTIONS: Full anthropometry, 7-day dietary assessment and physical activity levels were obtained. Insulin sensitivity (%S) and beta-cell function (%B) were determined using the homeostatic model assessment (HOMA). IMCL levels were determined using in vivo proton magnetic resonance spectroscopy; total body fat content was assessed by bioelectrical impedance. RESULTS: There was no difference between the groups in sex, age, BMI, waist measurement, percentage body fat, activity levels and energy intake. Vegans had a significantly lower systolic blood pressure (-11.0 mmHg, CI -20.6 to -1.3, $P=0.027$) and higher dietary intake of carbohydrate (10.7%, CI 6.8-14.5, $P<0.001$), nonstarch polysaccharides (20.7 g, CI 15.8-25.6, $P<0.001$) and polyunsaturated fat (2.8%, CI 1.0-4.6, $P=0.003$), with a significantly lower glycaemic index (-3.7, CI -6.7 to -0.7, $P=0.01$). Also, vegans had lower fasting plasma triacylglycerol (-0.7 mmol/l, CI -0.9 to -0.4, $P<0.001$) and glucose (-0.4 mmol/l, CI -0.7 to -0.09, $P=0.05$) concentrations. There was no significant difference in HOMA %S but there was with HOMA %B (32.1%, CI 10.3-53.9, $P=0.005$), while IMCL levels were significantly lower in the soleus muscle (-9.7, CI -16.2 to -3.3, $P=0.01$). CONCLUSION: Vegans have a food intake and a biochemical profile that will be expected to be cardioprotective, with lower IMCL accumulation and beta-cell protective.